UniLend

P2: Prototype and User Testing Plan

COSC2596-F01 Interaction Design and Testing
Instructor: Dr. Somang Nam

Group 7

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Submitted [TEAM: P2 due date]

Online copy. Student numbers are removed, and notes marked TEAM are still being completed.

Prototype: https://henish73.github.io/unilend-prototype/

1. Introduction and revised rationale

UniLend is a mobile prototype that helps students lend and borrow small items on campus: a phone charger before a lab, a calculator before a quiz, an adapter for a short presentation. These are short-term needs, so buying the item makes little sense, and another student nearby often has one unused in a backpack.

Existing services. UniLend is not a new idea. Libraries of things already lend useful objects to a community and have been studied as a form of sharing (Ameli, 2017). Library of Things in London lends household items for a daily fee, and the Toronto Tool Library's Sharing Depot lends general items to paying members. Student-focused platforms also exist. BorrowIt, at De La Salle University in the Philippines, verifies student identity and lets lenders set a rental price, and Need That, launched by university students in 2024, combines buying, selling, borrowing and requesting. These examples came from a short search rather than a full market review. Sources are listed after the references.

How UniLend differs. The services above mostly rely on fees, memberships or a central collection. UniLend is free and peer-to-peer, and it targets exchanges that last minutes or hours between verified students at safe public spots. Its main difference is the Pass-It-On step, which asks the person who was helped to help someone else, rather than only rewarding the lender.

Why this topic. We chose the topic because of the student need above. So far, that need rests on the team's own experience and has not been measured, so the report treats it as an assumption. Testing will show whether students can use the design, not how common the need is.

From P1 to P2. The table shows how this report answers the P1 feedback.

P1 feedback	What changed in P2
Rewards go to the lender directly, so it is a transaction, and Pay It Forward is not reflected.	A Pass-It-On step now follows every return. A chain counter shows how far the help has travelled, and both borrower and lender earn points.
The research question says non-monetary, but vouchers and gift cards are monetary.	The rewards are now a swappable layer. This report uses Variant B, which is P1 as written, on the instructor's advice. The research question has been reworded so it no longer says non-monetary.
"Nobody came up with this" is not accurate, and ease of prototyping is not a reason.	Those claims are gone. Existing services are named above, and the topic is justified by student need.
Videos and blogs are not scholarly sources.	The references are now scholarly sources and the course slides only.

Research question. How can a points system with optional campus perks encourage peer-to-peer item sharing while keeping returns accountable?

Theory and what remains assumed. The norm of reciprocity describes pressure to repay those who helped us (Gouldner, 1960). Paying help forward to a third person is a different and less dependable behaviour. In five experiments, Gray et al. (2014) found that equal treatment was paid forward in kind, but greed was paid forward more than generosity. For UniLend, this means a chain will not necessarily form by itself, which is why Pass-It-On has to be visible and tested. On rewards, a

meta-analysis found that tangible rewards can undermine intrinsic motivation while positive feedback can enhance it (Deci et al., 1999; see also Ryan & Deci, 2000). Gamification means using game design elements in non-game contexts (Deterding et al., 2011), and a review found mostly positive effects that depend on context and users (Hamari et al., 2014). People join collaborative consumption for reasons that include sustainability, enjoyment and economic gain (Hamari et al., 2016). **That rewards will raise lending on campus is an assumption, not evidence.** This study does not test it.

The one uncertainty this prototype tests. Can a student borrow, return and lend with no hints, and do they notice and use Pass-It-On? The rewards variant is not the uncertainty.

2. Requirements traceability

This table follows the chain from the course slides: persona need, scenario, storyboard moment, requirement, then screen. The personas are proto-personas, because the team has not carried out user research. The status column says what supports each row. Testing is the first evidence for any row marked as assumed.

The proto-personas are: **the rushed borrower**, a student between classes with something urgently missing; **the cautious lender**, who has spare items but is unsure about lending to a stranger; and **the helped student**, who has just been helped and could pass it on.

Persona need	Scenario	Storyboard moment	Requirement	Screen	Status
Rushed borrower needs a trusted group	First opens the app	Asked who can join	Limit access to verified campus members	1	P1 scope. Need assumed
Rushed borrower has little time	Phone at 5%, lab in an hour	Scans what is nearby	Let a student find a suitable item in seconds, by browsing or search	2, 3	Assumed
Rushed borrower wants to feel safe meeting	Agrees to meet a stranger	Picks where to meet	Only offer busy, public meeting places	4	P1 safety concern. Assumed
Both students need certainty	Meets at the library desk	"Did that count?"	Confirm each exchange to both sides, with visible feedback	5, 7	P1 accountability aim; slides on feedback
Rushed borrower needs a way forward	Item was just taken	Request cannot go through	Explain what happened and offer an alternative with no lost effort	12	Slides on recovery
Both students need a way forward	Code will not scan	Stuck at the meeting	Provide another way to confirm, and say nothing was recorded yet	12	Slides on recovery
Borrower needs to know what is owed	Finished with the charger	"When is it due?"	Show who, where and when for each loan, with one clear way to return it	6	Assumed
Helped student could help someone else	Has just returned the item	A chance to give back	Invite the helped person to help another, with an easy way to decline	8	P1 feedback. Effect assumed; tested in T2

Persona need	Scenario	Storyboard moment	Requirement	Screen	Status
Helped student wants to see the effect	Curious about the chain	Sees it grow	Show how far help has travelled	8, 11	P1 feedback. Effect assumed
Cautious lender wants control	A classmate asks for an item	Decides yes or no	Show who, what and when before deciding, and let the lender decline without giving a reason	9	Assumed
Cautious lender wants acknowledgement	Has just lent an item	"What did that get me?"	Show what was earned and allow one simple way to use it	10	Motivation assumed; not tested
Students want to see community effort	Between classes	Checks standing	Show collective impact and top helpers	11	P1 scope. Motivation assumed
Students use phones outdoors and on the move	Crossing campus in glare	Tapping with one hand	High contrast, large targets, works with screen readers	All	P1 commitment; slides on interface principles

3. Prototype building journey

From concept to testable design. P1 described points, a goodies shop and a leaderboard. The feedback showed that this rewarded lenders but gave borrowers no reason to pass help on. We added Pass-It-On to fix this, and made it the centre of the design.

The rewards-module decision. P1 mentioned gift cards while its research question said non-monetary. Instead of building around one reading, we separated the design into a reward-neutral core and a swappable rewards layer. Monetary content appears in only four places: the catalogue items, the leaderboard banner, the welcome line and the report wording. This report uses Variant B, which is P1 as written, on the instructor's advice. The research question has been reworded so it no longer says non-monetary. All participants will see one variant. We will not split the sample between versions, since a tiny classroom sample cannot pick a winner, and the slides name that as an A/B testing mistake.

The rewards used in this round. Points can be claimed for recognition, priority access and donations, plus vouchers, and top yearly lenders win gift-card prizes. The catalogue adds a campus coffee voucher and a bookstore discount to the recognition items.

Build choices. We built a coded mobile web prototype so that one setting (`rewards = recognition | perks`) controls the variant, and so that it opens on any phone or computer with no installation. The layout adapts to the screen: phones and tablets use bottom tabs, and desktops use a sidebar with multi-column pages. All sessions will use a phone, so that every participant sees the same layout. Because one participant has to finish a two-person exchange alone, a simulated student, Sam, replies a few seconds after each action, and the QR scan is a camera-style screen that completes on tap. All data is fictional.

Changes from the P1 scope. The prototype keeps P1's campus email check, categorised feed, points, past claims, leaderboard, safe pickup zones and dual QR scans. Three things are simplified. Points vary with item demand but not with borrowing time. The daily cap on points from one person is explained in

the help text but not simulated, because one participant cannot trigger it. Free access to paid academic websites is left out of the catalogue. These do not affect the one uncertainty being tested.

Checks and changes so far. We ran an automated walkthrough in a browser at phone size, covering all three tasks in both variants, the recovery states, undo, and the reset links. It also checks every screen for sideways scrolling, dead ends and for accessibility problems that automated rules can detect (WCAG 2 A and AA). All 128 checks pass in the current build, and the screen checks run at four sizes, from a small phone to a desktop, plus a phone in landscape. The shortest route takes 8 taps for T1 including sign-in, 3 taps to confirm the return in T2 (6 including Pass-It-On), and 7 taps for T3, which meets the target of about eight or fewer. These checks changed the prototype:

- The leaderboard banner was a different height in the two variants, which broke the fixed-slot rule. It now has a fixed height.
- A notification covered the Confirm hand-off button for a few seconds after Sam arrived, which would have looked like a participant error. Notifications now sit near the top and do not block taps.
- Confirmation text displayed item names awkwardly, such as "usb-c and lightning", so each item now has a plain short name.

We then audited the prototype and test materials against the brief and the course slides, which led to these changes:

- **Task 3 label.** A button read "See what you've earned", which copied the task wording and revealed the route. It now reads "Done", so finding Rewards is part of the task.
- **Task 2 timing.** The script could have ended the task before the Pass-It-On prompt appeared. The timer now stops at the return confirmation, while observation continues until the participant says they are done.
- **Pass-It-On coding.** "Yes or no" could not separate people who never saw the prompt from people who chose to skip it, so it now has three values.
- **Pass-It-On screen.** Two competing primary buttons became two equal choices.
- **Recovery and accountability.** We added an overdue loan, a refused request and a failed meeting, as the slides ask for feedback and recovery beyond the happy path.
- **Post-test questions.** Two hypothetical questions now ask about past behaviour.
- **Other changes.** The main action now stays on screen on a phone in landscape, an offline notice was added, and participants can no longer reach the facilitator page from the app.

To make the sessions themselves more reliable, we then added four things:

- **A session log.** The app records each task run on the test phone, and the facilitator page exports it as a CSV file.
- **A facilitator panel.** Holding the UniLend name opens it on the participant's phone, so the team can reset between tasks without typing a web address.
- **Visible recognition.** Variant A's research question is about recognition, but claiming a badge used to change nothing on screen. A claimed badge now appears on the leaderboard and Kindness Wall, and passing help on adds you to the wall.
- **An offline copy.** The app is now stored on the phone, so a reload still works if the classroom Wi-Fi drops.

Last, we redesigned the interface to make it easier to find your way, without changing any task's steps or tap count. The home screen now opens with a "Right now" section for whatever needs attention. Each borrow, return and lend flow shows which step you are on and what comes next. Lists are grouped onto one surface, with fewer boxes, following the slides' advice to avoid rectangles within rectangles. The item page shows who is lending and the key facts at a glance, a small campus map highlights the chosen pickup spot, and the Rewards screen shows progress toward the next reward. We also added a dark mode, which follows the phone's setting or can be chosen under Profile and help, and every screen passes the same contrast checks in dark mode. All automated checks were run again after these changes.

These checks show that the flows work. They do not show that people find them easy, which only the sessions can tell us.

Pilot. [TEAM: after the practice session with someone outside the group, record what happened and what changed. Delete this note if the pilot comes after submission, and say it is still planned.]

Use of AI tools. As the course allows, Claude, an AI assistant from Anthropic, generated the prototype code and helped draft this report from the group's plan. Appendix A lists the prompts, and Appendix B lists what the AI assumed and how the group responded. The group is responsible for the content.

4. System overview

Only the bottom layer differs between variants, so every task and metric works on both.

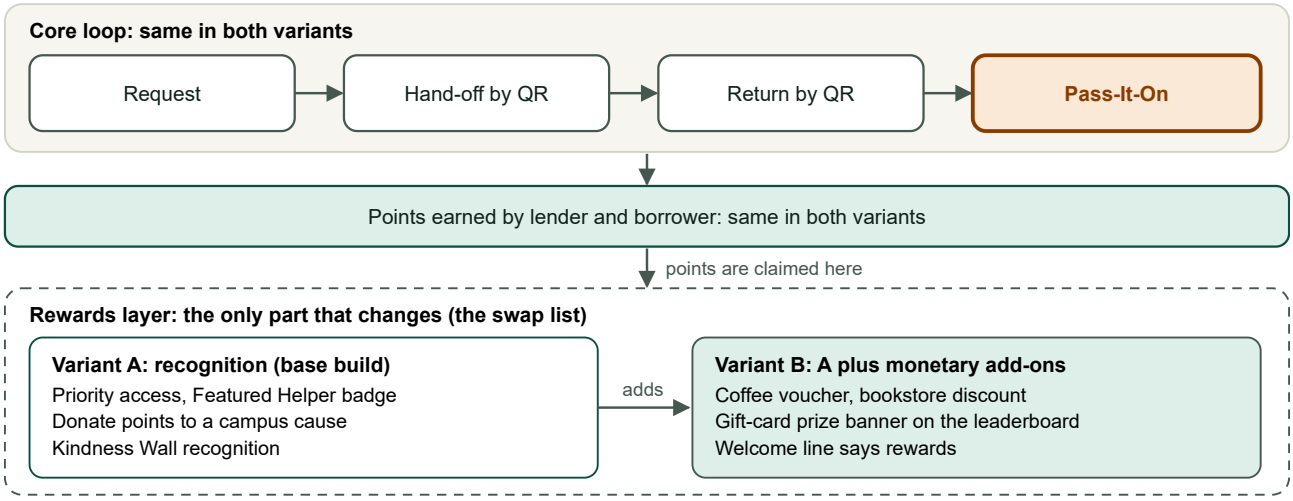
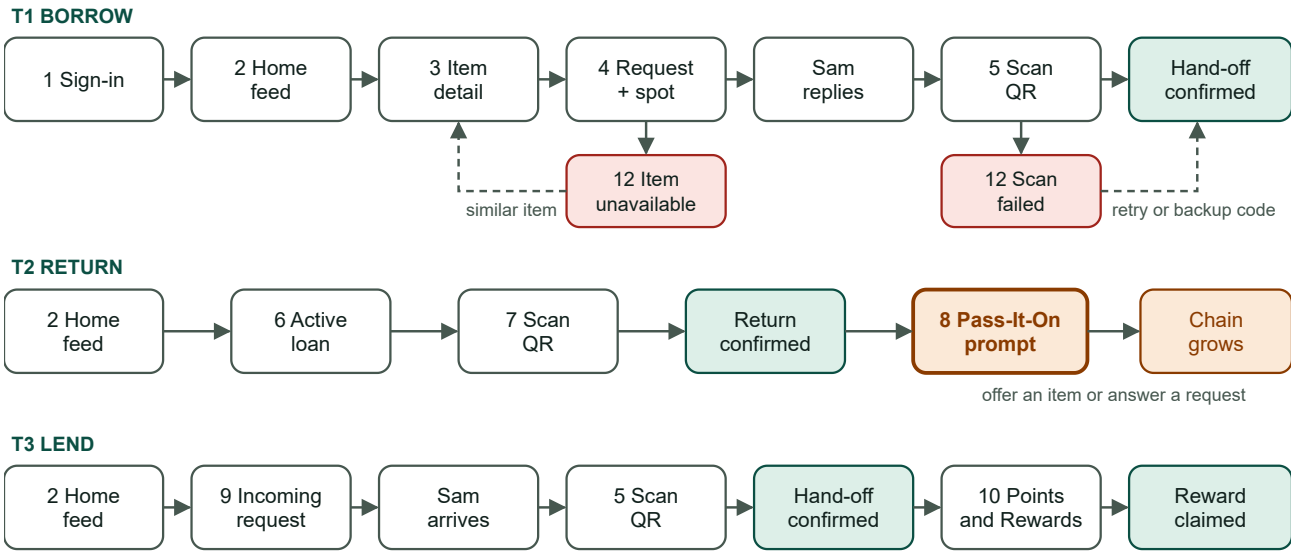


Figure 1. Design layers. The core loop and the points never change.



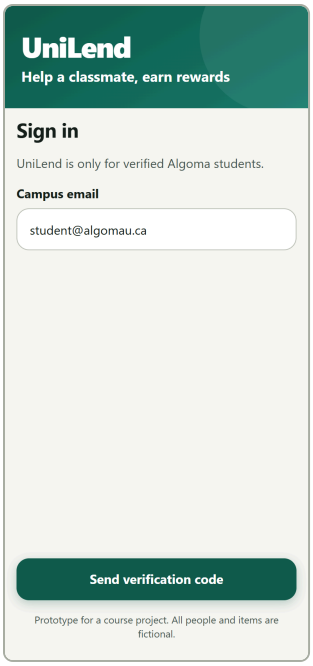
Green: a task's end state. Red: recovery states. Screen 11 (Community Impact), Loans and Rewards are always reachable from the bottom tabs.

Figure 2. Flow of the three test tasks. Back and undo are available throughout, so every path can be reversed.

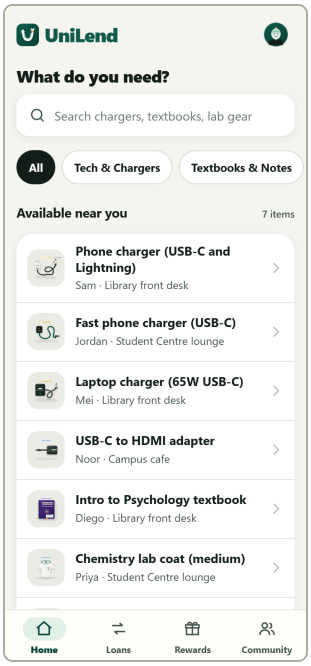
Screens

#	Screen	Tasks	Depends on rewards variant?
1	Sign-in with campus email (simulated verification)	All	Welcome line only
2	Home feed with categories and search	T1, T3	No
3	Item detail	T1	No
4	Request and pickup-zone choice	T1	No
5	Hand-off confirmation (simulated QR scan)	T1, T3	No
6	Active loan with return option	T2	No
7	Return confirmation (simulated QR scan)	T2	No
8	Pass-It-On prompt and chain counter	T2	No
9	Incoming request (lender view)	T3	No
10	Points and Rewards	T3	Catalogue content only
11	Community Impact and leaderboard	T3	Prize banner only
12	Recovery states: item no longer available, scan failed	T1, T2	No

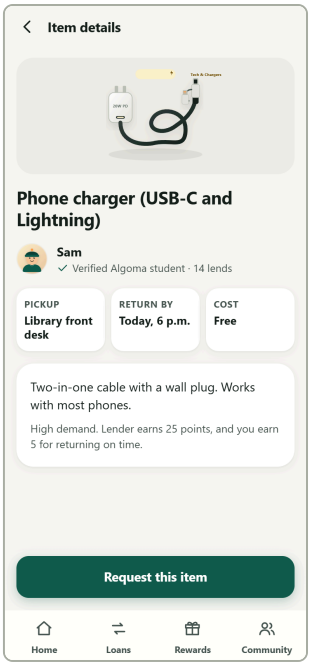
A Loans tab and a Profile and Help page support these screens. The Pass-It-On step reads "Someone helped you. Who can you help next?" and offers two ways to act: offer one of your own items, or answer an open request. It can also be skipped. Beyond screen 12, the prototype also handles a request that gets refused, a meeting that fails, and an overdue loan with a reminder, a request for more time and no fine. These support the accountability aim from P1, and are outside the three test tasks.



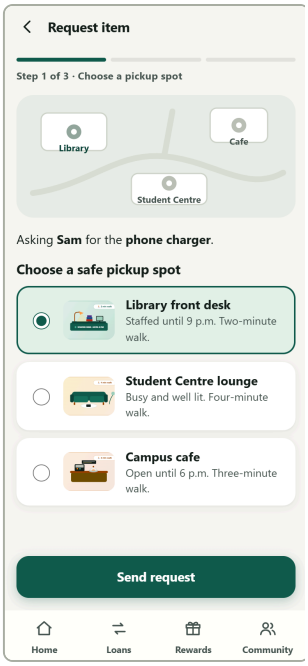
1. Sign-in



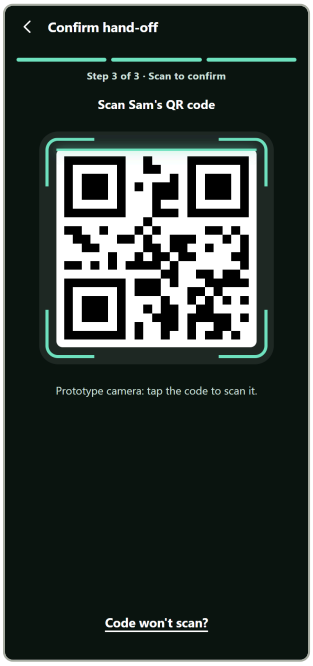
2. Home feed



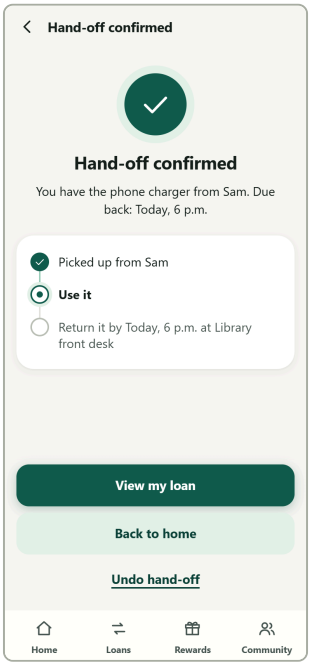
3. Item detail



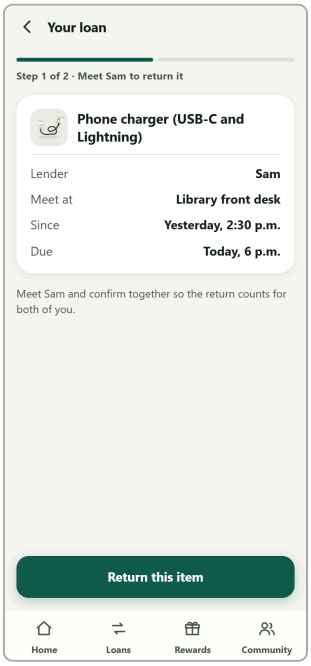
4. Request and pickup spot



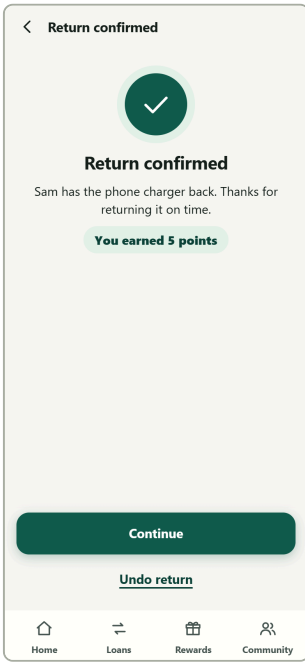
5. Hand-off scan



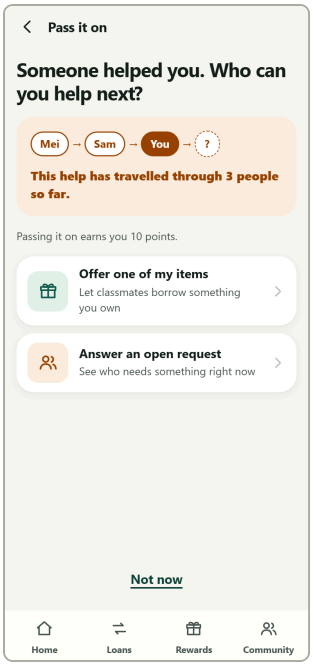
5. Hand-off confirmed



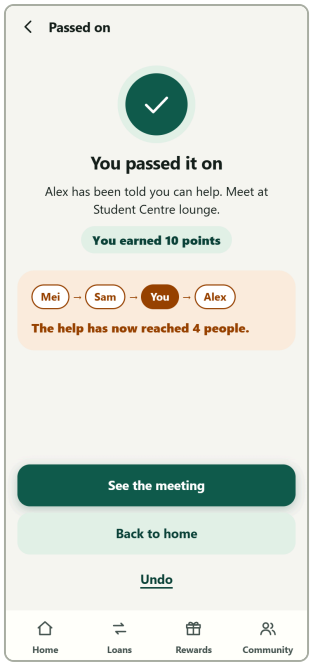
6. Active loan



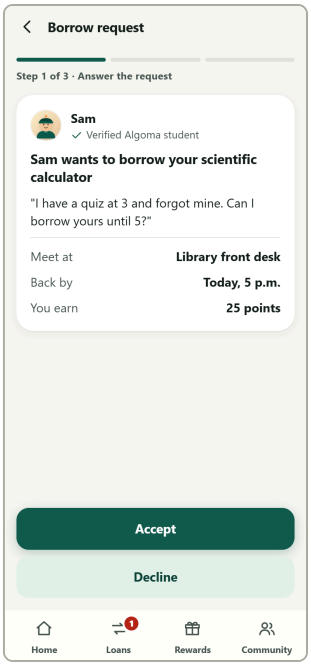
7. Return confirmed



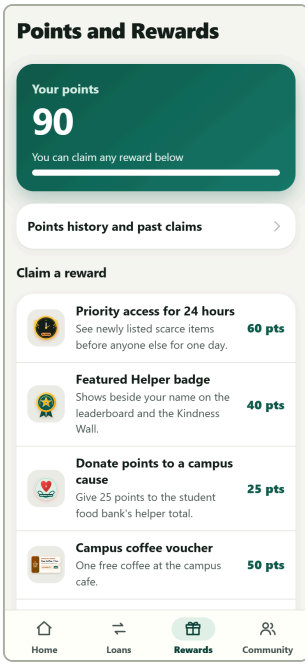
8. Pass-It-On prompt



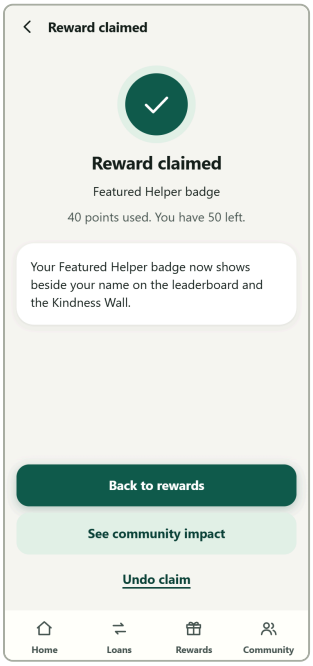
8. Chain counter grows



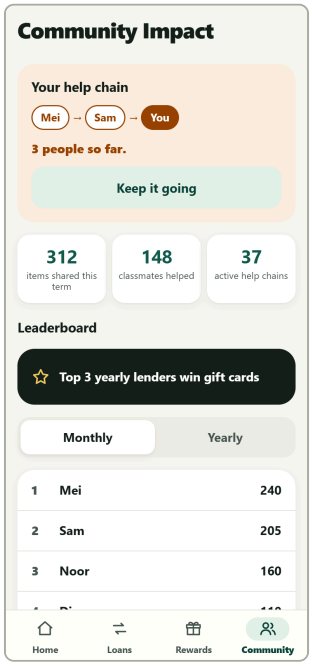
9. Incoming request



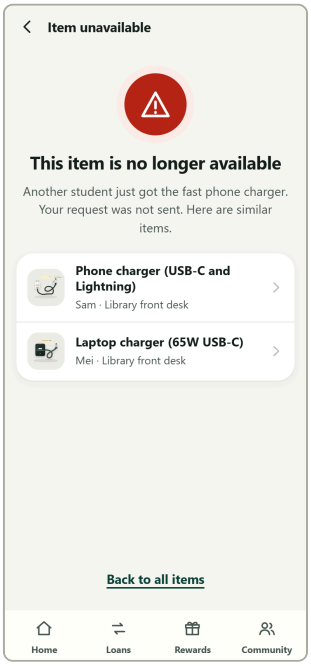
10. Points and Rewards



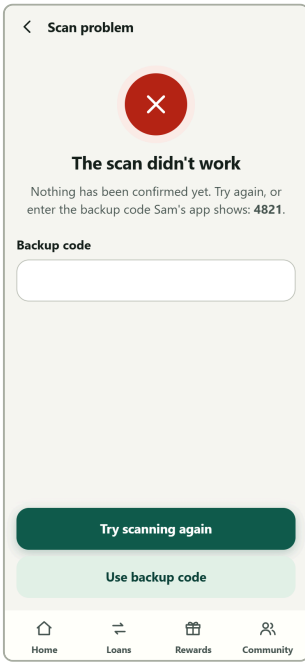
10. Reward claimed



11. Community Impact



12. Item unavailable



12. Scan failed

Figure 3. The prototype, Variant B (adds monetary perks), captured at phone size.

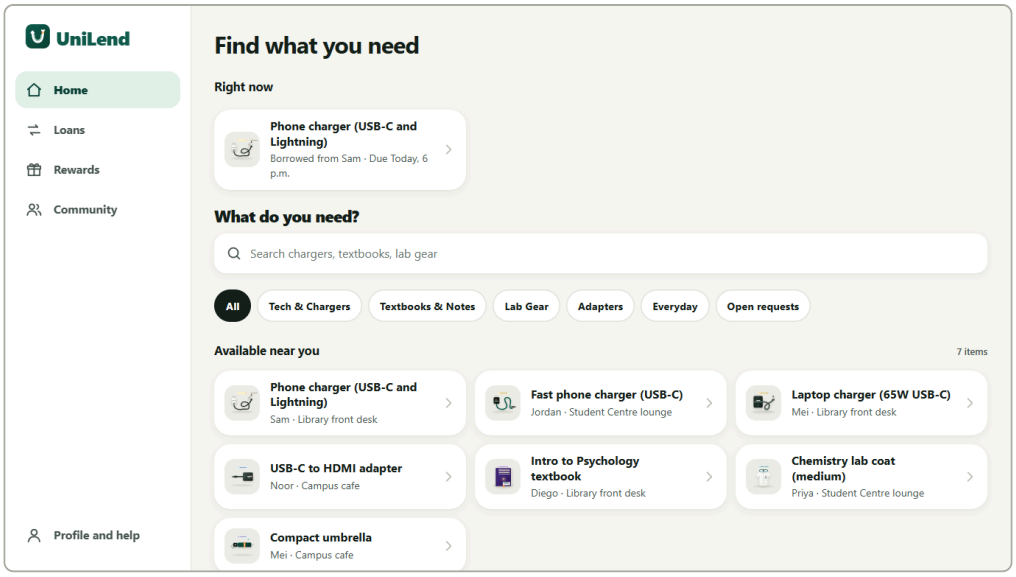


Figure 4. The same home screen on a desktop.

The swap list

Component	Variant A (recognition only)	Variant B (adds monetary)
Rewards catalogue	Priority access for 24 hours, Featured Helper badge, donate points to a campus cause	Everything in A, plus a coffee voucher and a bookstore discount
Leaderboard banner	"Top helpers are recognised on the Kindness Wall"	"Top 3 yearly lenders win gift cards"
Welcome line	"Help a classmate, earn recognition"	"Help a classmate, earn rewards"
Report wording	Section 1 and Limitations, A column	Section 1 and Limitations, B column

The catalogue items are placeholders for the team's final choices. Claiming takes the same steps and lands in the same place in both variants.

Testing support and limits

A separate facilitator page resets the prototype to each task's starting state, holds the task wording, and exports the session log. On the participant's phone, holding the UniLend name for about a second opens a small panel. There, the facilitator chooses the participant ID and task, which loads that task's starting state and shows the scenario to read aloud. Tapping Begin starts the log's timer. A participant is unlikely to trigger this by accident, but the team should watch for it. The prototype has no server, real camera, real email or real messaging. Each phone keeps its own state and log, so the team should use one phone for all sessions. Once the app has been opened on that phone, it keeps working without a connection. Set its Appearance to Light, so every participant sees the same design with the high contrast P1 promised for outdoor use.

5. Participants and recruitment

We will recruit 10 Algoma students plus 2 reserves.

Inclusion: currently enrolled at Algoma, 18 or older, owns a smartphone, and has borrowed or lent an item with another student at least once or says they would.
Exclusion: members of this group, and anyone who has already seen the prototype.

Demographic	Target mix (of 10)
Year level	At least 2 in first or second year, and at least 2 in third year or later
Phone	At least 3 iOS users and at least 3 Android users
Lending experience	At least two people who have never lent anything
Access needs and language	Noted at screening, so the setup can be adjusted. Anyone who needs an adjustment stays eligible

Recruitment. Each group member will invite students they do not work with closely, through classes, residence and student common areas, and will share the screener. Testing day may also give access to students in other groups, who qualify if they have not seen the prototype. We will keep filling slots until 10 are booked and 2 reserves are named. Taking part is voluntary, with no payment. [TEAM: confirm these channels are realistic.]

> **Invitation.** "Hi, our group in COSC 2596 is testing the design of a phone app for lending items between students, and we're looking for participants. It takes about 20 minutes on campus on [date]. We're testing the design, not you, and there's no payment. You need to be an Algoma student aged 18 or older with a smartphone. Could you answer five short questions so we can book a time?"

No-shows. We will send a reminder the day before. If someone has not arrived 5 minutes after their start time, the next reserve takes the slot. If both reserves are used, we will book more sessions on campus until 10 are complete. Anyone who stops partway stays in the data with what they completed, and does not count toward the 10 unless all three tasks were attempted.

Screener

1. Are you currently enrolled at Algoma University?
2. Which year are you in?
3. Which phone do you use?
4. In the past term, have you borrowed or lent an item (charger, calculator, textbook) with another student?
5. Is there anything we should adjust so you can use a phone app comfortably?

Before booking, we also confirm that the person is 18 or older and has not seen the prototype.

Sessions and schedule

Each session takes about 20 minutes. This is an estimate we will adjust to the time available.

Part	Minutes
Welcome, consent, permission to record	3
Pre-test background questions	2

Part	Minutes
Three tasks, up to 3 minutes each	9
Post-test questions	4
Reset and notes	2
Total	**20**

Roles. The facilitator reads the script and stays neutral. The note-taker records observations and keeps time. The floater resets the prototype and tracks the schedule. With five group members, we can run two stations at once with one shared floater, which allows all 10 sessions in about 100 minutes.

Round	Start (from first session)	Station 1	Station 2
1	0:00	P01	P02
2	0:20	P03	P04
3	0:40	P05	P06
4	1:00	P07	P08
5	1:20	P09	P10
Reserve	1:40	P11	P12

Testing is planned for the class testing day, which the latest course slides list as November 16. [TEAM: confirm this date and the room, assign people to each role, and add clock times.] Any sessions that do not fit on the day will be booked on campus in the days that follow.

Team roles for the project. [TEAM: name who owns prototype build, test materials, report and references, and video and data sheet.]

6. Test materials

Informed consent form

UniLend usability study: COSC 2596, Algoma University

Purpose. Group 7 is testing the design of a prototype app for lending items between students. We are evaluating the design, not you.

What you will do. Answer a few background questions, try three short tasks on a phone while saying what you are thinking, and answer a few questions afterwards. It takes about 20 minutes.

Voluntary. Taking part is your choice. You may pause or stop at any time, or skip any question, with no consequence.

Rewards in the app. Rewards shown in the app are fictional and no real prizes are given. You will not be paid for taking part.

Risks and benefits. Risks are minimal, and no more than everyday phone use. There is no direct benefit to you, but your input will improve the design.

Your data. We record your actions, comments and answers under a participant number, not your name. This includes an automatic log of the taps you make on the test phone. Short quotes may appear in our course report and presentation with no name attached. Recordings and this form are kept by the group and seen only by the group and the course instructor. They will be deleted when the course ends.

Questions. Contact [TEAM: group contact email] or the course instructor, Dr. Somang Nam.

Please tick to agree:
☐ I am 18 or older and I agree to take part.
☐ I agree to have the phone screen and my voice recorded. (You can take part without this.)

Name _____ Signature _____ Date _____

Pre-test background questions

1. What program and year are you in?
2. How comfortable are you using apps on your phone? (1 not at all to 5 very)
3. Tell me about the last time you needed something on campus that you did not have with you. What did you do?
4. Have you ever lent something to another student? What was that like?
5. Have you used any app or service for borrowing or sharing items? Which one?

Facilitator script

> **Opening (read aloud).** "Thanks for coming. We are evaluating the design. Any difficulties help us improve it. You may pause or stop at any time. Please describe what you are looking for and expecting, as you go. [Name the observers.] May we record the phone screen and your voice?"
>
> "The rewards shown in the app are fictional, and no real prizes are given. Please focus on what you are trying to do, not on the colours or fonts. I'll read you a situation, and you decide what to do. I can't tell you how to do things, because we want to see how the app works on its own. Tell me when you think you're done."
>
> **For each task.** Hold the UniLend name to open the panel, then choose the participant and the task. Read the scenario word for word, tap Begin, and hand the phone over. Start the stopwatch at the same moment. Stop timing at the task's confirmation, when the participant says they are done, or at 3 minutes. In Task 2, stop timing at "Return confirmed", then stay silent and keep observing until the participant says they are done. Then ask: "How easy was this task, from 1, very difficult, to 7, very easy?" and "How did that task go for you?"

Neutral prompts only. Use these if the participant stops talking or gets stuck:

Do not say	Say
The blue button will let you continue.	What would you do next?
That was easy, right?	How did that task go for you?
You missed the menu.	What are you looking for?
Why didn't you click that?	What did you expect that option to do?

- **Hints.** Anything that reveals the route counts as a hint and is logged. If a participant is stuck for about a minute and asks for help, first ask a neutral prompt. A hint is a last resort, and that task is then scored as assisted.
- **Pass-It-On in T2.** Say nothing about it, before or during the task. Record *taken* if they use it, *declined* if they see it and skip it, and *not seen* if they say they are done before reaching it.
- **Visual comments.** Do not engage. Log comments on colours or fonts separately, then say: "Thanks. Please carry on with what you were doing."
- **Stopping.** Stop at the time limit, or if the participant asks. Reassure them: "That helps us. It tells us something to fix in the design."

Close (read aloud). "That's all the tasks. Thank you. Remember that the rewards in the app were fictional. Please don't describe the app to other students until testing is finished, because we need people to see it fresh."

Post-test questions

1. "The rewards on offer made me more likely to help someone." (1 strongly disagree, 2 disagree, 3 neither agree nor disagree, 4 agree, 5 strongly agree) Why?
2. After you returned the charger, what did you see, and what did you make of it?
3. Which moment was most confusing or frustrating?

4. Think of the last time you needed something on campus that you did not have. How did what you just did compare with what you did then?
5. Think of the last time you lent something to another student, or chose not to. What decided it? Did anything you saw today relate to that?
6. Is there anything else you would like to tell us?

Question 1 measures opinion, not lending behaviour, and the results will be described that way.

7. Tasks

The three tasks cover borrowing, returning and lending. None names a reward type or a screen, and each gives a realistic goal rather than the route. The time limit for each is 3 minutes, or until the participant says they are done.

	Task 1: Borrow an item	Task 2: Return an item and see what follows	Task 3: Lend an item and look at what you have earned
Scenario (read aloud)	"It's 2 p.m., your phone is at 5% and you have a lab at 3. Get a charger from another student on campus."	"You borrowed a charger yesterday and are done with it. Give it back."	"A classmate has asked to borrow your scientific calculator. Help them out. Then see what your help has earned you and do something with it."
Starting state	Sign-in screen, campus email filled in	Signed in, charger on loan from Sam	Signed in, Sam's request waiting
Success, with no hints	A pickup spot is chosen and the hand-off is confirmed.	The return is confirmed.	The request is accepted, the hand-off is confirmed, and one claim is completed on the Rewards screen. Any catalogue item counts.
Observed, not required	Which charger is chosen; any recovery from a taken item	Whether Pass-It-On is taken, declined or not seen, with no prompting, and what they said	Whether they find Rewards on their own; which item they claim
Shortest route (facilitator only)	8 taps: send code, verify, item, request, spot, send, confirm hand-off, scan	3 taps to confirm: loan, return, scan. Then continue and two taps to pass it on	7 taps: request, accept, confirm hand-off, scan, Rewards tab, item, claim

Each task is scored by the same rule in both variants, so the criteria do not change with the rewards decision. Before the pilot, we will check that every task has a reachable end state and that each reset link returns to the right starting state.

8. Data framework

We capture the same measures for every participant and task, and tag each row with the variant shown. None of the measures depends on reward type.

Metric	Type	How it is measured	Tasks
Task completion	Quantitative	Success, assisted or fail, by the success rule in section 7	T1-T3
Time on task	Quantitative	Seconds from the end of the scenario to the task's confirmation, using a timer or the screen recording. In T2 this is the return confirmation. In T1, sign-in time is also recorded separately, so T1 can be compared with the other tasks	T1-T3
Errors	Quantitative	Count of wrong taps that need a backtrack, plus recoveries from the failed-scan and unavailable-item states	T1-T3
Hints given	Quantitative	Count of facilitator prompts that reveal the route. The target is zero	T1-T3
Ease rating	Quantitative	One question after each task: "How easy was this task?" on a 1-7 scale	T1-T3
Pass-It-On	Behavioural	Taken, declined or not seen, observed in T2 without prompting	T2
Recovery state seen	Behavioural	None, item, scan or both. These states depend on the path taken, so not everyone meets them	T1-T3
Observed behaviour	Qualitative	Notes in the slide's format: participant and task, then what they did	T1-T3
Think-aloud quotes	Qualitative	Short verbatim quotes, kept as usability comments or logged apart as visual comments	T1-T3
Perceived motivation	Rating and open text	A single Likert item, post-test: "The rewards on offer made me more likely to help someone", 1-5, plus "Why?"	After T3
Issue severity	Qualitative	Each problem found is rated 0-4 by the team after the session (Nielsen, 1994)	All

Outcome coding. Success means the end state was reached with no hints. Assisted means it was reached after a hint. Fail means it was not reached in the time limit or the participant gave up. A participant who says "easy" but needed two hints is reported as assisted, along with their rating. **Severity:** 0 not a problem, 1 cosmetic, 2 minor, 3 major, 4 catastrophe.

Data sheet

The workbook has one row per participant per task. This example shows the format and is not a result:

```
participant_id,variant,task,outcome,time_sec,errors,hints,ease_1_7,pass_it_on_taken,observed_behaviour,quote,severity_0_4,sign_in_sec,recovery_seen
P01,A,T1,success,74,1,0,6,,Opened Help first then found the feed,"I wasn't sure where to start",2,12,none
```

Session log. The prototype also logs each task run: time from Begin to the task's confirmation, taps and backtracks up to that point, recovery states seen, and the Pass-It-On outcome. The note-taker's sheet stays the main record, because the log cannot tell a mistaken tap from someone exploring, or see a hint. We will use the log to check the stopwatch times and the Pass-It-On coding. Where the two disagree, the screen recording decides. The exported rows go in the workbook's "Auto log" sheet.

Other sheets hold screener answers, post-test answers and visual comments, and a summary sheet calculates the results by task. The issue log follows the slides: observation, interpretation, revision, then follow-up. For example, the format is "3 of 5 participants searched Help before finding bookings"

(observation), "the label may be unclear" (interpretation), "try a clearer entry label" (revision), "test the revised path with new participants" (follow-up).

Analysis for P3. We will report completion rates, mean times, errors, hints and ease by task, plus the shares who saw and took Pass-It-On, and group the qualitative notes into themes. With 10 participants, these results are descriptive. They can reveal usability problems, but they cannot show that rewards raise lending.

Limitations

- Cash-value rewards may reduce intrinsic motivation, and funding for prizes is unconfirmed.
- The exchange is simulated. Participants never meet a real stranger or hand over their own property, so the test measures usability, not real willingness to lend.
- A prompt in a test may be followed more readily than in real life, and one session cannot show whether chains last. Being observed can also change behaviour (the Hawthorne effect), which may raise the Pass-It-On rate.
- The tasks run in a fixed order, because sign-in comes first. Later tasks may look easier because participants have learned the app.
- A convenience sample at one university may not represent other campuses, and hi-fi screens can draw comments on visual style. The script and the separate visual-comments log address the second issue.

Dry-run video and next steps

A proxy who did not build the prototype will complete the three tasks with this script in one unedited take, targeting about 2.5 minutes. The prototype and rewards variant are frozen before recording. The 10 sessions then take place during the evaluation sprint, and the results feed the analysis for P3.

9. References

Ameli, N. (2017). Libraries of Things as a new form of sharing. Pushing the sharing economy. *The Design Journal, 20*(sup1), S3294-S3304. https://doi.org/10.1080/14606925.2017.1352833

Deci, E. L., Koestner, R., & Ryan, R. M. (1999). A meta-analytic review of experiments examining the effects of extrinsic rewards on intrinsic motivation. *Psychological Bulletin, 125*(6), 627-668. https://doi.org/10.1037/0033-2909.125.6.627

Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). From game design elements to gamefulness: Defining "gamification". In *Proceedings of the 15th International Academic MindTrek Conference* (pp. 9-15). ACM. https://doi.org/10.1145/2181037.2181040

Gouldner, A. W. (1960). The norm of reciprocity: A preliminary statement. *American Sociological Review, 25*(2), 161-178. https://doi.org/10.2307/2092623

Gray, K., Ward, A. F., & Norton, M. I. (2014). Paying it forward: Generalized reciprocity and the limits of generosity. *Journal of Experimental Psychology: General, 143*(1), 247-254. https://doi.org/10.1037/a0031047

Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does gamification work? A literature review of empirical studies on gamification. In *Proceedings of the 47th Hawaii International Conference on System Sciences* (pp. 3025-3034). IEEE. https://doi.org/10.1109/HICSS.2014.377

Hamari, J., Sjöklint, M., & Ukkonen, A. (2016). The sharing economy: Why people participate in collaborative consumption. *Journal of the Association for Information Science and Technology, 67*(9), 2047-2059. https://doi.org/10.1002/asi.23552

Nam, S. (2026). *COSC 2596 Interaction Design and Testing: Lectures 1-8* [Lecture slides]. Algoma University.

Nielsen, J. (1994). Heuristic evaluation. In J. Nielsen & R. L. Mack (Eds.), *Usability inspection methods* (pp. 25-62). John Wiley & Sons.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist, 55*(1), 68-78. https://doi.org/10.1037/0003-066X.55.1.68

Existing services named in section 1 (examples only, not cited as evidence)

BorrowIt: https://thelasallian.com/2020/10/15/borrowit-an-item-sharing-platform-by-lasallians-for-lasallians/ · Need That: https://www.thedp.com/article/2024/11/penn-startup-apps-trading-items · Library of Things: https://www.ealing.gov.uk/a_to_z/service/649/library_of_things · Toronto Sharing Depot: https://amp.cbc.ca/lite/story/1.3496027

Appendix A. AI use and prompts

The course allows AI use if it is disclosed with the prompts, and if the group can defend its decisions. Claude Code (Claude Opus 5.5, Anthropic) was used for this project. It was given the group's plan, the P2 brief, P1, the P1 feedback and the lecture slides as input.

#	Prompt (as typed)	What it produced
1	"Can you implement the whole plan and build me what's required?" (with the group's plan PDF attached)	First build of the prototype, the data sheet, the video plan, the draft message and the first draft of this report
2	"Help me host it", followed by replies while publishing, and "can you do github instead"	Published the prototype to GitHub Pages and tested the live link
3	"Can you make website is both mobilr optimised and web optimized both, i want overall improvement of UI."	Responsive layouts for phone, tablet and desktop, and visual refinements
4	"Can you audit and suggest some more improvements aligning project requirements & also lecture notes."	An audit against the brief and slides, with ranked findings
5	"Yes, apply them"	The audit fixes listed in section 3
6	"Can you suggest some more improvements for protype" and "Build items 1-4"	The session log, facilitator panel, visible recognition and offline copy
7	"Can you change whole UI/UX, make it wonderful & Very easy to navigate."	The redesign described in section 3
8	"Can you also make dark mode as well."	The dark mode and the Appearance setting
9	"can you also get assets live."	Added the existing illustrations and icons to the app, fixed a broken icon and clipped text, and published the documents page

[TEAM: add the prompts used for anything else, including the group's plan if AI helped write it, and any AI use by other group members.]

Appendix B. What the AI assumed, and our review

The course slides ask which user needs AI invented, and what still needs confirming. These details came from the AI, not from research or the group. Each one needs a decision from the group.

AI-supplied detail	Where it appears	Group's decision
Who the proto-personas are, and what they need	Section 2	[TEAM: evidence or changes]
Point values (25, 15, 10, 5) and reward costs	Prototype, screens 8-11	[TEAM: keep or change]
People, items, walking times, opening hours and leaderboard numbers (fictional)	Prototype	[TEAM: keep or change]
Pass-It-On shown as two equal choices, and the wording of its options	Screen 8	[TEAM: keep or change]

AI-supplied detail	Where it appears	Group's decision
Late returns: a reminder, a request for more time, no on-time points	Prototype, overdue state	[TEAM: keep or change]
Two-station schedule and recruitment channels	Section 5	[TEAM: confirm they are realistic]
Who can see the data, and deletion when the course ends	Consent form	[TEAM: confirm, or check with the instructor]
Choice of existing services and summaries of sources	Sections 1 and 9	[TEAM: read the sources and confirm]
How the log defines a backtrack (back, undo, cancel) and when its timer starts (at Begin)	Section 8	[TEAM: keep or change]

What we changed or rejected from the AI's output, and why: [TEAM: the group writes this in its own words. The course slides do not allow taking AI output as is.]